Regularization for Neural Networks

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Data Preprocessing

- Mean subtraction
- Normalization
- PCA and whitening

Normalization

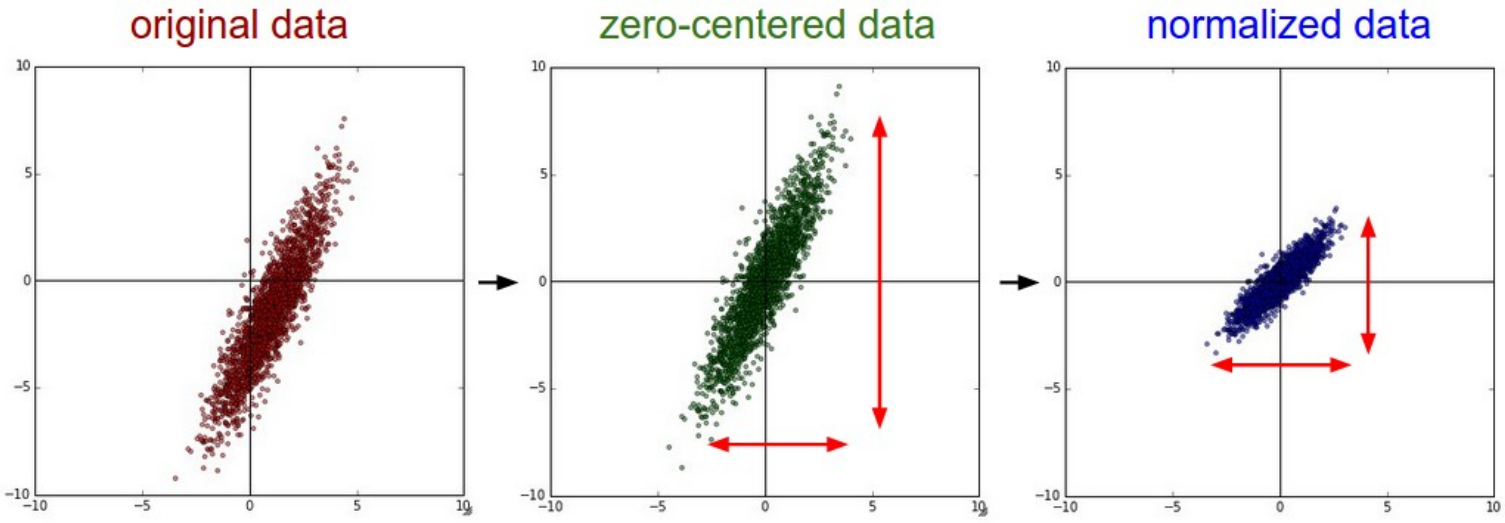


Figure 1: http://cs231n.github.io/convolutional-networks/

PCA and Whitening

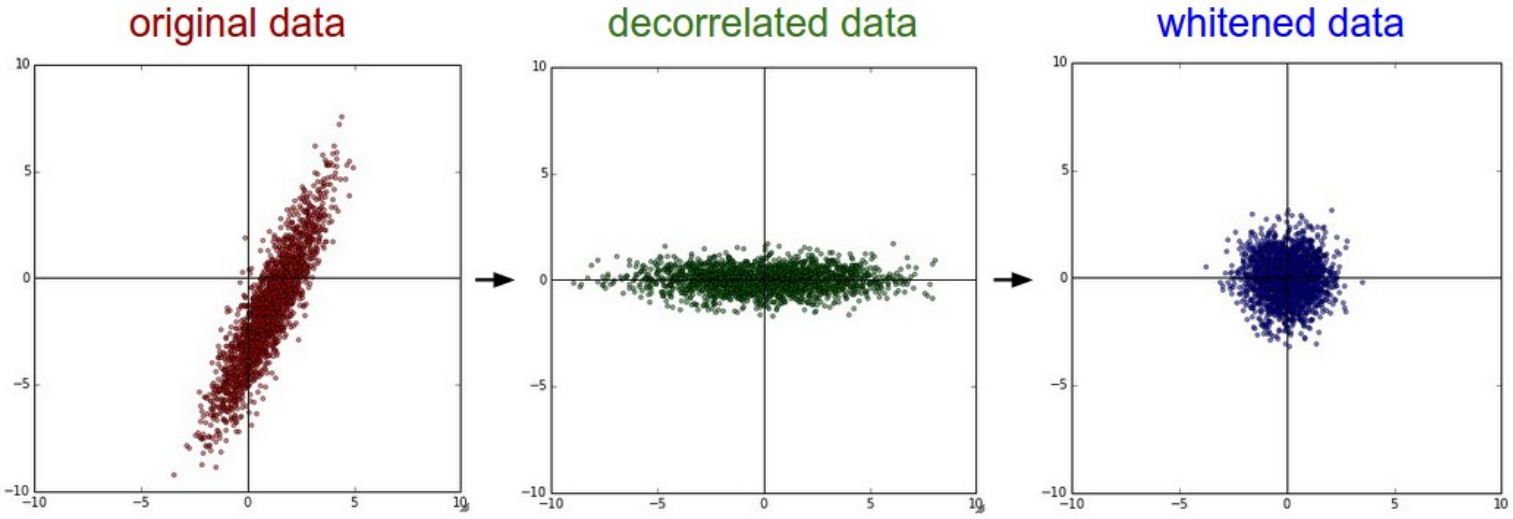


Figure 2: `http://cs231n.github.io/convolutional-networks/`

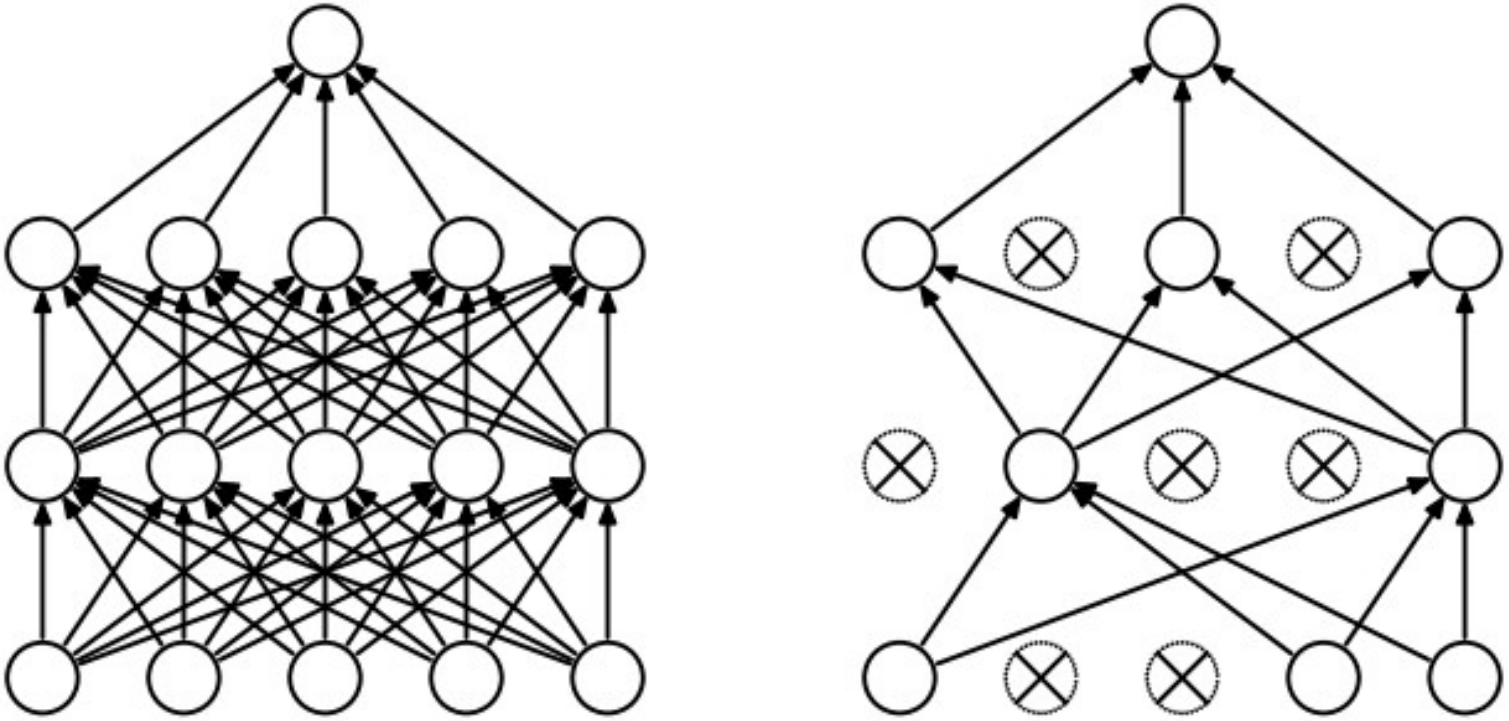
Weight Initialization

- All zero initialization
- Small random numbers
- Calibrating the variances with $1/\sqrt{n}$
- Sparse initialization
- Rectified linear unit (ReLU)
- Batch Normalization

Regularization

- l_2 regularization
- l_1 regularization
- Max norm constraints
- Dropout

Dropout



(a) Standard Neural Net (b) After applying dropout.

Figure 3: http://cs231n.github.io/convolutional-networks/

References

- Professor Seungjin Choi's materials
- http://cs231n.github.io/convolutional-networks/

Thank you.